PAPER_088: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

Title: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57, [SCm] ≈ 0.99)
Date: March 7, 2026
Source Data: MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule), validate_phase3.py
Index Slot: 1.11 Black Hole Physics & Hawking Radiation,

Title: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57, [SCm] ≈ 0.99)
Date: March 7, 2026
Source Data: MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule), validate_phase3.py
Index Slot: 1.11 Black Hole Physics & Hawking Radiation, PAPER_088

Abstract

The neutrino spectral energy distribution (SED) from a black hole system particularly Sagittarius A* probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission (T_UQFF = 0.99 T_H), AGN corona pp/p? interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of MAIN_1_CoAnQi.cpp implements the `NeutrinoSEDModule` returning flux predictions at E_? = 1 TeV10 PeV for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by f_TRZ = 0.01.

UQFF Discovery: Novel application of UQFF calibration constants (? = $5.0\times10^{-}4$ day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
Hawking emission	T_UQFF = 0.99 T_H	E_? T_H ~ 10?4 K (negligible)	Sub-dominant
Corona pp/p?	Ug4 AGN feedback	1 TeV 10 PeV	Dominant
TRZ vacuum	f_TRZ = 0.01	0.1×100 PeV	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** (T_H ~ 10?4 K ? E_? ~ k_B T_H « 1 eV). The interesting band is TeVPeV cosmic neutrinos from AGN corona interactions.

---

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity L_corona = 10? L_Edd:

$$\Phi_\nu^{\rm SM}(E_\nu) = \Phi_0 \left(\frac{E_\nu}{\rm TeV}\right)^{-\gamma} e^{-E_\nu/E_{\rm cut}}$$

With ? = 2.2, E_cut = 5 PeV.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor f_TRZ = 0.01 enhances corona pair production:

$$\Phi_\nu^{\rm UQFF}(E_\nu) = \Phi_\nu^{\rm SM}(E_\nu) \times (1 + f_{\rm TRZ}) = 1.01 \times \Phi_\nu^{\rm SM}(E_\nu)$$

2.3 Ug4-Mediated Enhancement

The Ug4 energy density (3.352941 × 10 J/m baseline) additionally contributes to pion production:

$$\Delta\Phi_\nu^{Ug4}(E_\nu) = f_{Ug4} \cdot \Phi_\nu^{\rm SM} \cdot \frac{U_{g4}}{U_{g4,\rm ref}}$$

For current Sgr A* quiescent state (A_AGN = 1, e^{-?t} near-unity): f_Ug4 « f_TRZ.

---

3. Multi-Flavor Ratio

Standard model predictions: ?*e : ?* : ?_t = 1:2:0 at production ? 1:1:1 after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_\mu : \nu_\tau)_{\rm UQFF} = (1:1:1) \times (1 + 0.001\, f_{\rm TRZ})$$

The 26D mixing is degenerate in flavor at the detector level **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

---

4. NeutrinoSEDModule (Batch 21)

From MAIN_1_CoAnQi.cpp Batch 21:

```
// namespace SOURCE_BATCH21
// class NeutrinoSEDUQFFTerm : public PhysicsTerm
double NeutrinoSEDUQFFTerm::compute() const {
    // Returns integrated flux: E^2 dF/dE at E = E_ref (1 TeV)
    double SM_flux = phi0 * pow(E_ref/TeV, -gamma) * exp(-E_ref/E_cut);
    double TRZ_factor = 1.0 + f_TRZ;  // = 1.01
    double Ug4_factor = 1.0 + f_Ug4 * Ug4_current / Ug4_ref;
```

```
    return SM_flux * TRZ_factor * Ug4_factor;
}
```

---

5. validate_phase3.py Validation

`validate_phase3.py` performs phase-3 cross-validation of Batch 21 outputs against `uqff_validation_test.py`:

- **Neutrino SED test:** UQFF flux = 2s deviation from IceCube diffuse background ? **PASS**
- **Multi-flavor test:** (1:1:1) ratio ? **PASS**
- **TRZ enhancement:** ? = 1.0% above SM ? **PASS**
- **Ug4 quiescent:** negligible Ug4 contribution at A_AGN = 1 ? **PASS**

---

6. IceCube-Gen2 Detectability

Current IceCube sensitivity: F_3s 10? TeV cm? s^{-1} sr? for Sgr A* point source.

UQFF prediction: ? = 0.3% excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state (A_AGN » 10) would push Ug4 enhancement to ~5% potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 (1.001)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
AGN-active Ug4	+0.35%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule) | validate_phase3.py | f_TRZ = 0.01 | IceCube constraints

---

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0×10^{-4} day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor

Symbol	Value	Description
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
$-\Sigma\lambda_i \cdot U_i \cdot$E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times \left(1 + 10^{13}\,\Theta(\rho_{SCm} - \rho_c)\right) \times \left(1 + A_q \cos(\Delta\omega\, t)\right)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds

Mode	Dominant Term	Primary Use Case
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.